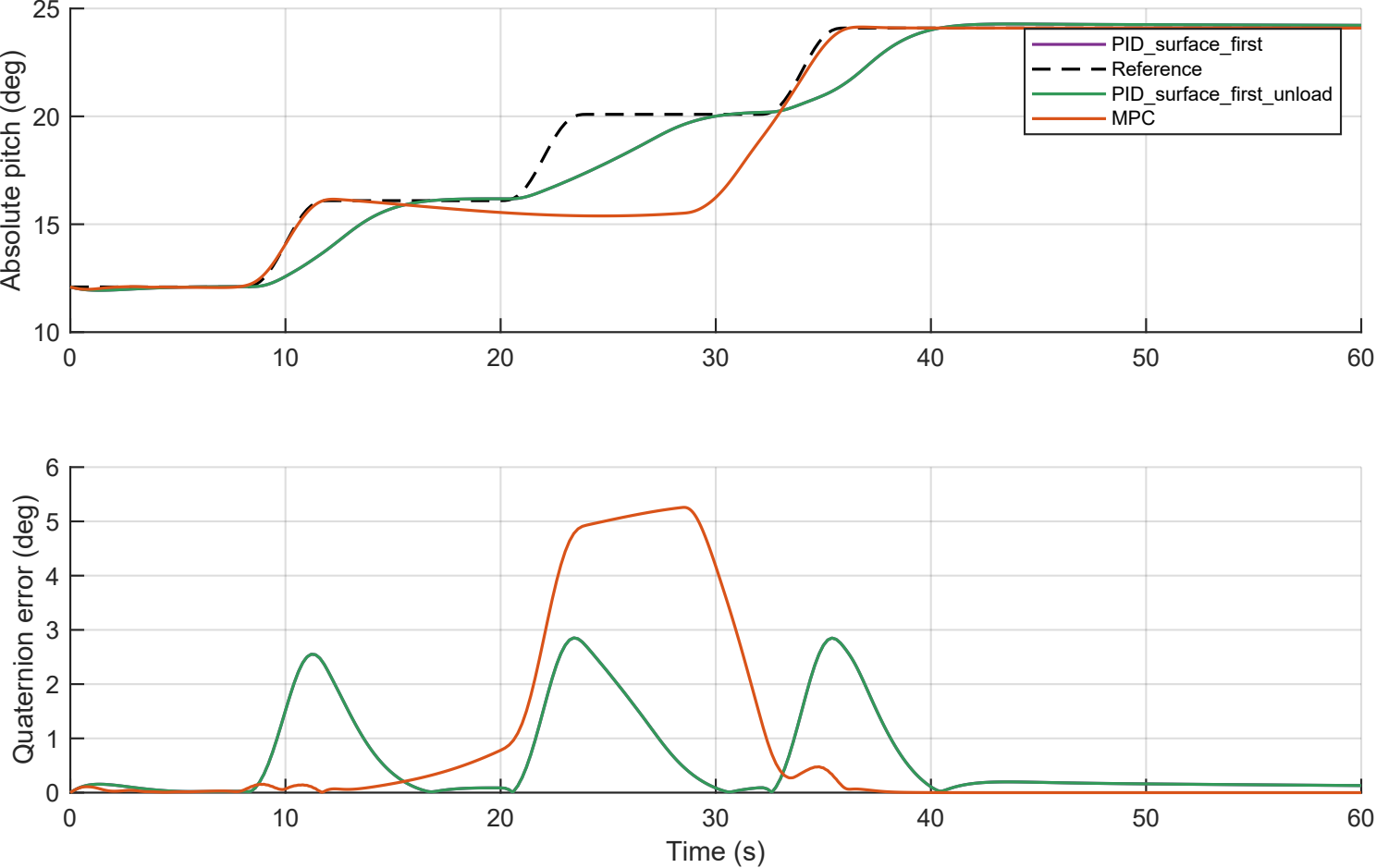

Absolute pitch (deg)
25
20
15
10
0
10
20
30
40
50
60
PID_surface_first
Reference
PID_surface_first_unload
MPC
Quaternion error (deg)
6
5
4
3
2
1
0
Time (s)